

TARBI PYAKUREL

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EDUCATION

Bachelor of Science in Computer Science | *The University of Southern Mississippi*

Expected May 2027

Minor: Economic Data Analysis | Cumulative GPA: 3.90/4.00

Selected coursework: Algorithms, Artificial Intelligence, Mathematical Statistics, Linear Algebra, Programming Languages, Operating Systems, Computer Networks, Database Design, Web Application Security

RESEARCH EXPERIENCE

Undergraduate Research Collaborator — Multilingual LLM Safety | *Independent*

2026–Present

collaboration with a PhD researcher

- Investigating jailbreak behavior and emergent misalignment in large language models, with emphasis on Nepali as a low-resource language.
- Developing categories of Nepali-language prompts and documenting cases in which language or prompt formulation changes model safety behavior.
- Working toward a reproducible evaluation protocol that separates preliminary observations from validated findings and supports comparison across models.
- Considering responsible handling of adversarial prompts, human evaluation by Nepali speakers, and quantitative measures of attack success and response safety.

Undergraduate Research Assistant — MCCPO Reinforcement Learning | *Collaboration*

Summer 2026

with a graduate researcher

- Assisted in formulating an adversarial environment for evaluating the behavior and robustness of the MCCPO reinforcement-learning algorithm.
- Contributed to the experimental framing by considering how environmental conditions and adversarial behavior could expose limitations in the learning procedure.
- Gained experience translating a research question into an evaluation setting and distinguishing exploratory observations from defensible experimental conclusions.

RESEARCH INTERESTS

Trustworthy AI; multilingual and low-resource NLP; LLM safety and alignment; adversarial machine learning; reinforcement learning; secure AI-enabled systems

TECHNICAL PROJECTS

Continuing Education Learning Management System | *Startup production application*

- Built and launched a production LMS serving more than 20 users in its first month and processing live payments through Stripe.
- Designed a Next.js, TypeScript, PostgreSQL, and Prisma architecture with secure authentication, role-based administration, media delivery, file uploads, and automated certificates.
- Deployed the platform with Docker Compose, automated database migrations, and environment-based configuration; iterated features using user feedback.

QuietHelp — Anonymous Peer-Support Platform | *Next.js, Spring Boot, FastAPI, Redis,*

PostgreSQL, Docker

- Engineered a distributed microservices platform with real-time WebSocket communication and AI-assisted support.
- Designed an API gateway with rate limiting and service routing and implemented fallback behavior for failures involving external AI services.

MyShop — E-Commerce Platform | *Spring Boot, MySQL, Redis, Docker*

- Built microservices for authentication, products, carts, payments, and machine-learning-based recommendations.
- Implemented JWT authentication, role-based access control, Redis token management, and containerized deployment.

LEADERSHIP AND TEACHING

Co-Lead | *Google Developer Group on Campus*

Feb 2026–Present

- Helped organize developer workshops, RoboSoccer activities, technical watch parties, and campus events; represented the group at Google I/O and HackPrinceton.

Mathematics Learning Assistant | *The University of Southern Mississippi*

Aug 2025–Present

- Explain mathematical concepts in group settings and help students develop structured analytical problem-solving skills.

International Student Ambassador | *The University of Southern Mississippi*

Jan 2024–Present

- Mentor incoming international students and coordinate orientation activities with university faculty and staff.

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, SQL

Frameworks: Spring Boot, FastAPI, React, Next.js

Data and infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Docker, Git, GitHub, AWS, CI/CD